



Midnight Tuning

Extended XDFs - Advanced HEGO Mode Setup

Getting Started with Advanced HEGO Mode

Thanks to advancements with definition version 1.3, it is now possible to enable the EEC via a J3 chip to interpret the signal from a wideband oxygen sensor. Using a wideband over a narrowband carries many benefits, such as more consistent and controllable fueling in closed loop, enabling closed loop beyond lambda targets of 1.0 (Closed loop at WOT) and enabling the use of modern, faster and more responsive sensors.

High Level Overview

This patch works by enabling the EEC to use a table to define the HEGO “Switchpoint” as opposed to a single value. While you could continue to use the single value with the wideband sensor, the table enables “biasing” the air fuel ratio target under closed loop depending on RPM and MAP to enable running the engine more favorably in different conditions, unlike with a narrowband which must run at or very close to 1.000 lambda.

In order to get a wideband signal to the EEC, this patch is designed to take advantage of the Techedge Wideband product, which possesses a special feature enabling the voltage on its o2 sensor output pins to be adjusted however the user sees fit. This can be used to make the wideband signal fit inside the 0 to 1.665 volt range of the oxygen sensor input on the EEC.

Step 1 - Hardware Setup (Wiring)

The first step in this patch is to install the required hardware in the car. You'll need a techedge wideband with an NBSim pin (The popular 2J9 is supported and recommended, but the more advanced units like the 3Ax and 3Ex are also supported) or any wideband that enables customisation of its analog o2 signal outputs. This guide only covers the techedge, which can be purchased alongside a J3 chip, programmer, etc from Ti Performance.

Install the oxygen sensor as normal, replacing the factory narrowband as you will no longer need it. Run the cable for the oxygen sensor back into the interior to the techedge, which you should place next to or near the ECU. To wire the techedge, I recommend following the below image guide to see the pin numbers on the EEC, you will need to pull off the black cover on the EEC connector to reveal the pin numbers on the connector and it is also recommended to peel back some of the loom tape on the ECU harness to reveal more of the wiring and make it more accessible for tapping into.

² Shift Solenoid 2 - (SS2)	1	OUTPUT	OUTPUT 53	-(SS3) - Shift Solenoid 3 ²
Engine Enable Input - (EEI)	2	INPUT	OUTPUT 54	-(GSCD) - LPG Stepper Motor D ¹
Heated Exhaust Gas Oxygen Sensor Ground - (HO2SG)	3	INPUT	SYSPWR 55	-(KAPWR) - Keep Alive (Memory) Power
⁶ Air Conditioner Pressure Transducer - (ACPT)	4	INPUT	56	
⁵ Traction Control Engine Torque Output - (MMI)	5	INPUT	INPUT 57	-(KS) - Knock Sensor Positive
Camshaft Position Sensor Negative - (CMP)	6	OUTPUT	INPUT 58	-(VSS) - Vehicle Speed Sensor
⁶ Air Conditioner Refrigerant Discharge Temp. - (ACRT)	7		INPUT 59	-(CCS) - Cruise Control Signal ⁴
	8	INPUT	60	
	9	INPUT	61	
	10		INPUT 62	-(MMR) - Traction Control Engine Torque Request ⁵
² Power Economy Switch - (PES)	11	INPUT	INPUT 63	-(FSS) - Fuel Select Switch ¹
¹ LPG Stepper Motor C - (GSCC)	12	OUTPUT	INPUT 64	-(TR) - Transmission Range Sensor ²
EEPROM Power Supply - (FEPS)	13	SYSTEM	65	
	14		66	
BUS Negative - (BUSN)	15	SYSTEM	OUTPUT 67	-(EVAP) - Evap Canister Purge Solenoid
BUS Positive - (BUSP)	16	SYSTEM	68	
Electro-Drive Fan 1 - (EDF1)	17	OUTPUT	INPUT 69	-(ACR) - Air Conditioning Relay ⁶
	18		70	
Starter Relay - (START)	19	OUTPUT	SYSPWR 71	-(VPWR) - Vehicle Power
¹ LPG Stepper Motor A - (GSCA)	20	OUTPUT	OUTPUT 72	-(VCT) - Variable Camshaft Timing ³ or (GCO) Gas Cut Out ¹
Crankshaft Position Sensor Positive - (CKP+)	21	INPUT	OUTPUT 73	-(INJ5) - Injector 5
Crankshaft Position Sensor Negative - (CKP-)	22	INPUT	OUTPUT 74	-(INJ3) - Injector 3
	23		OUTPUT 75	-(INJ1) - Injector 1
Power Ground - (PWRGND)	24	SYSPWR	SYSPWR 76	-(PWRGND) - Power Ground
Case Ground - (CASEGN)	25	SYSPWR	SYSPWR 77	-(PWRGND) - Power Ground
Ignition Coil 1 - (COIL1)	26	OUTPUT	OUTPUT 78	-(COIL3) - Ignition Coil 3
² Shift Solenoid 1 - (SS1)	27	OUTPUT	79	
² Shift Solenoid 4 - (SS4)	28	OUTPUT	OUTPUT 80	-(FP) - Fuel Pump
	29		OUTPUT 81	-(VPS) - Variable Pressure Solenoid 5 ²
⁶ Air Conditioner Evaporator Temperature - (AEAT)	30	INPUT	82	
Power Steering Pressure Switch - (PSP)	31	INPUT	OUTPUT 83	-(IAC) - Idle Air Control Solenoid
Knock Sensor Negative - (KS)	32	INPUT	84	
	33		INPUT 85	-(CMP) - Camshaft Position Sensor Positive
Temp. & Manifold Absolute Pressure [Press]- (T-MAP)	34	INPUT	INPUT 86	-(BFS) - Blower Fan Speed ⁶
Heated Exhaust Gas Oxygen Sensor - (HO2S)	35	INPUT	87	
	36		88	
² Transmission Fluid Temperature Sensor - (TFT)	37	INPUT	INPUT 89	-(TP) - Throttle Position Sensor
Engine Coolant Temperature Sensor - (ECT)	38	INPUT	SYSPWR 90	-(VREF) - Vehicle Reference Voltage
Temp. & Manifold Absolute Pressure [Temp]- (T-MAP)	39	INPUT	INPUT 91	-(SIGRTN) - Signal Return
	40		INPUT 92	-(ACD) - Air Conditioner Demand ⁶
¹ Gas Tank Level Sensor - (GLVL)	41	INPUT	OUTPUT 93	-(RCO) - Air Conditioning Recirculation Override ⁶
Electro-Drive Fan 2 - (EDF2)	42	OUTPUT	OUTPUT 94	-(PCS) - Power Control, Solenoid 6 ²
Data Output Link - (DOL)	43	SYSTEM	95	
	44		OUTPUT 96	-(LUS) - Lock Up, Solenoid 7 ²
Clean Tachometer Output - (CTO)	45	OUTPUT	SYSPWR 97	-(VPWR) - Vehicle Power
	46		OUTPUT 98	-(IMRC) - Intake Manifold Runner Control
	47		OUTPUT 99	-(INJ6) - Injector 6
¹ LPG Stepper Motor B - (GSCB)	48	OUTPUT	OUTPUT 100	-(INJ4) - Injector 4
	49		OUTPUT 101	-(INJ2) - Injector 2
	50		102	
Power Ground - (PWRGND)	51	SYSPWR	INPUT 103	-(GLVL-RTN) - Gas Tank Level Sensor Return ¹
Ignition Coil 2 - (COIL2)	52	OUTPUT	104	

(AU Intech Series 2+3 pinout shown here, if you are doing this mod on a V8 or E series, use another diagram to suit your exact car)

To connect the Techedge, first connect the Power / Positive on the 2 pin power ground connector of the wideband to EEC pin 71 (VPWR), this is ignition key power to the EEC and should be connected to the positive power connection of the Techedge, strip back some insulation from the wire and make a solder connection, its not recommended to cut the wire to make the connection. Then connect the Techedge Ground on the 2 pin power connector to pin 76 or 77, either pin works as both are ECU Power Ground join this the same way as the power wire, its not recommended to cut the wire to make the connection, just strip some insulation upstream and make a solder connection.

The last connection to make is NBsim, with some length of wire from the EEC left, cut the wire running to Pin 35 and connect the wire left from the EEC to the NBsim pin of the Techedge, the other end runs to the old o2 plug in the engine bay and can be left disconnected.

Step 2 - Techedge Setup (Sensor Calibration)

With the techedge installed, you can check its working by turning the ignition on and verifying the controller powers on, leave the o2 sensor disconnected for this and just verify that the lights come on. If all is well with the controller, you can move to sensor calibration.

With the oxygen sensor connected, connect the USB / Serial connector from the techedge to your laptop, and run WBUtil (Download it here: <https://wbo2.com/sw/util.htm>). Once your USB / serial cable is connected, turn on the ignition with the oxygen sensor connected but not installed in the exhaust, make sure not to let the end of the sensor touch anything, it should be surrounded by pure air, it will also get very hot (Don't touch).

In WBUtil on the General Page shown after starting the app, pick the Com Port associated with the Wideband (Hopefully there may only be one shown). You can then check the Terminal tab and look for activity, if there is activity, you should be connected and ready to go.

Go to the calibrate tab and press Start to begin automatic sensor calibration, once the blue dot in the green box is overtop the red dot (8192 also shown in green next to it in the lpx box) press STOP to conclude the calibration, your now ready to install the sensor in the exhaust and the wideband is ready to operate.

(Refer to the Ti Performance Documentation for more details on Techedge Setup, this is only a basic guide)

Lastly, we need to configure NBsim for the J3 patch. Go to the WB Tables tab and select NBsim PWM under the output pin. Under Linear Data Points, Set Rich AFR to 10, Lean AFR to 20, Rich Volts to 0 and Lean Volts to 1.67. Then click Generate Table, and then WRITE to program the Wideband.

All done! The techedge is set up and ready to work, as well as provide a wideband signal to the EEC. You can move onto the J3 setup.

Step 3 - J3 Chip / Tune Setup

Once your Hardware is installed and Techedge Configured, you are ready to start tuning with a wideband directly connected to the EEC. Make these changes first to get up and running.

Firstly, Head to Bin Components > Advanced O2 and Wideband Closed Loop Control

- Apply the Advanced HEGO Mode patch, to enable the CL target table.
- In the Patched EGO Switch Point vs RPM and MAP (Techedge Lambda) Table, initially set all the cells to 1.000 to target 14.7 for now. You can tune this later to whatever AFR / Lambda you like.
- Under EGO Switch Point Hysteresis, lower this substantially, 20 Counts works well, but if you find the EEC gets confused about the engine being rich or lean (Trimming the wrong way or getting lost) you can increase this value some more.
- Lastly, under Invert EGO Input, be sure this is unchecked (Stock is checked). This value tells the EEC if 0 volts corresponds to being rich or lean and the stock sensor works "backwards". If you fail to check this, the EEC will misinterpret rich as lean and trim fuel the wrong way. You will know quickly if you got this wrong.

This is the essential changes, but there are some other changes you should consider making under Fueling > HEGO and Adaptive Learning.

- Under HEGO Gain, you can reduce this substantially to reduce oscillation / hunting caused by the sensor response time creating overshoot. This couples nicely with disabling jumpback (Talked about further down the page). Really you should tune this to your liking, it can be varied based on MAP and RPM.
- Under HEGO Bias, you can fill this table with 0. Otherwise the EEC has a tendency to stay on the richer or leaner side of the swinging motion longer (This feature is intended to offset AFR with a narrowband).
- Under both HEGO Jumpback Multiplier functions, multiply the Jumpback Mul values by 0.25 to greatly reduce the jumpback effect. This effect is designed to cause the old narrowband sensor no longer installed to switch the other way, without the HEGO Gain having to start trimming fuel in the other direction, but this is no longer really required.
- If you intended to run Closed Loop even at WOT, or run substantially leaner or richer than 1.0 lambda in closed loop, you should consider disabling adaptive learning. The long term fuel trim is actually an airflow multiplier, so leaving this enabled may cause a stacking effect with your base fuel tables when not in closed loop. Use the Min / Max ECT to enable to achieve this.

All Done!!!

Now your ready to go out and drive your newly wideband converted EEC car! Fine tune as you see fit and enjoy the modern wideband fueling control.

For more Information on Ti Performance / General J3s:

[Visit Ti Performance](#)

For more Information on Midnight Tuning Supplied J3s & EEC Tuning:

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